Week-1 Documentation

CPU & Instruction Execution.

This Topic which tells about the how CPU connected into the STC89C52 microcontroller and also what is the process happened in between while execution of the instructions.in the system instructions which means tells to the system to perform this kind of things which is pre-instructed into the code.

Module Integration

The simulator is divided into interconnected modules. The Process Manager manages multiple programs using PCBs and the ready queue. The Scheduler selects the next process using FCFS, Round Robin, or priority scheduling and performs context switching. The selected process is executed by the CPU, which interacts with Memory and Stack during instruction execution. The GPIO, Timer, and Interrupt modules provide peripheral functionality. The UI integrates all modules and displays the current system and performance metrics.

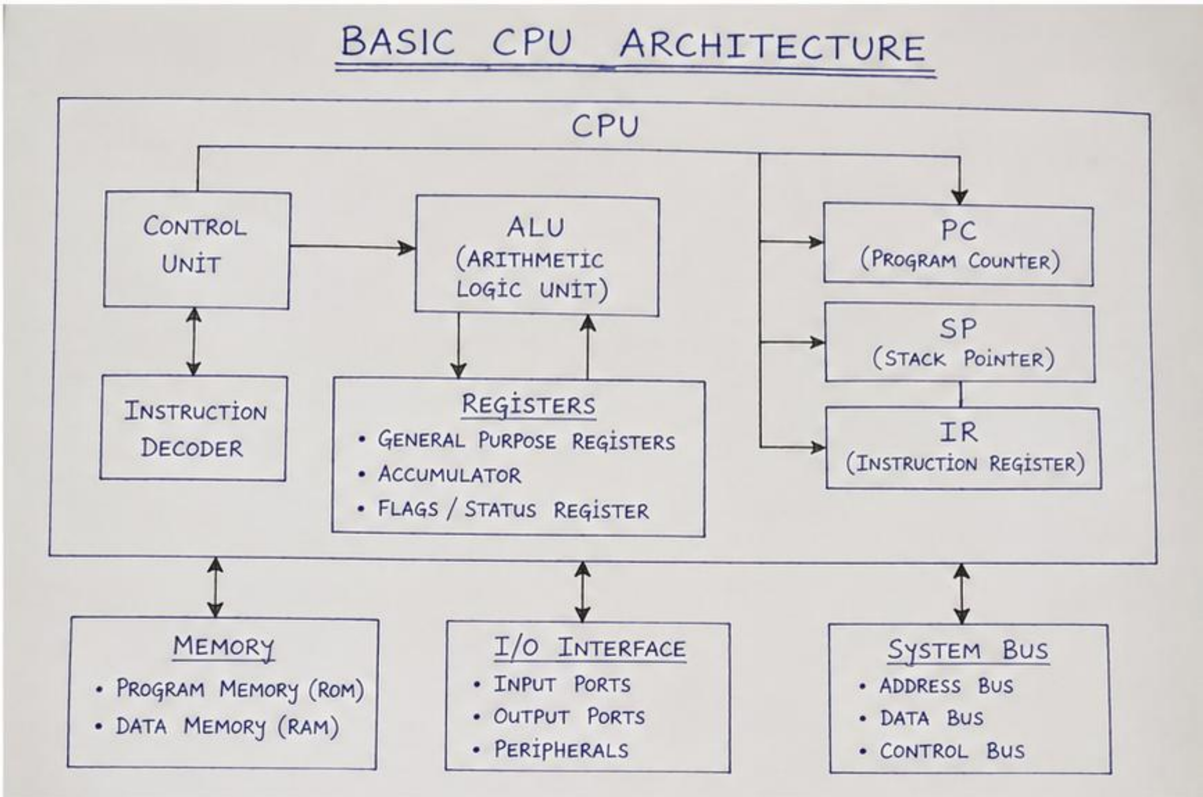
<u>CPU Architecture</u>

The CPU is the brain of the microcontroller, it receives instructions, understands them, performs the required task, and produce the result.

Fetch ⟶ Decode ⟶ Execute

Main Components of CPU

- ALU (Arithmetic and Logical Units)
- Control Unit
- Registers
- Program Counter
- Stack Pointer
- Instruction Register
- Status/Flag Registers



STC89C52 Architecture

The STC89C52 is an 8-bit microcontroller based on the 8051 architecture it integrates a CPU program memory, data memory ,I/O ports ,timers/counters, serial communication and interrupt control on a signal chip. The microcontroller executes instructions stored in program memory and uses its internal registers and peripherals to perform different operations.

Simulators main features which is taken by the its datasheet.

- 8-bit 8051-compatible CPU
- Program memory capacity
- Internal RAM capacity
- Four 8-bit I/O ports
- Timers/counters
- Serial communication interface
- Interrupt system
- Oscillator/clock system

Main Registers.

Accumulator(ACC): The accumulator commonly called register A is an 8-bit register used extensively during arithmetic ,logical and data-transfer operations. Many instructions use the accumulator as the destination for the result.

Example: MOV A, #25H

B Register: The B register is an 8-bit register mainly used along with the accumulator for multiplication and division operations. It can also be used as a general-purpose register in other operations.

Example:

MUL AB

DIV AB

PSW -Program Status word: The PSW is a special register that contains status information and control bits related to CPU operations. It includes important flags such as carry ,auxiliary carry and overflow, along with register-bank selection bits.

R0-R7 Registers: R0-R7 are eight general -purpose registers available in the selected register bank they are used to temporarily store data during program execution. The 8051 architecture provides multiple registers banks, with the active bank selected through the PSW.

Program Counter (PC): The PC is used to keep track of the address of the next instruction to be fetched from program memory. After an instruction is executed ,the PC normally advances to the next instruction, while jump and branch instructions can change its value.

Stack Pointer(SP): The Stack Pointer is an 8-bit register that points to the current top of the stack in internal RAM. It is used during stack operations such as PUSH and POP and is also involved when subroutines and interrupts require stack storage.

Data Pointer(DPTR): DPTR is a 16-bit register formed by DPH and DPL. It is used mainly for addressing data and for certain data transfer operation.

Main memories

- Program Memory
- Data Memory

I/O Ports: The STC89C52 provides four 8-bit I/O ports, P0, P1, P2 and P3. These ports allow the CPU to communicate with external devices. Some port pins also have alternate functions, depending on the microcontroller configuration.

Interrupt System: An **interrupt system** is a mechanism in a microprocessor or microcontroller that allows the CPU to **temporarily stop its current program and respond to an important event**. Instead of the CPU continuously checking every device, a device can send an **interrupt signal** to the CPU when it needs attention.

Interrupts in 8051/STC89C52

The 8051-compatible architecture provides **five interrupt sources**:

Interrupt Source	
INT0	External Interrupt 0
Timer 0	Timer/Counter 0 overflow
INT1	External Interrupt 1
Timer 1	Timer/Counter 1 overflow
Serial	Serial communication

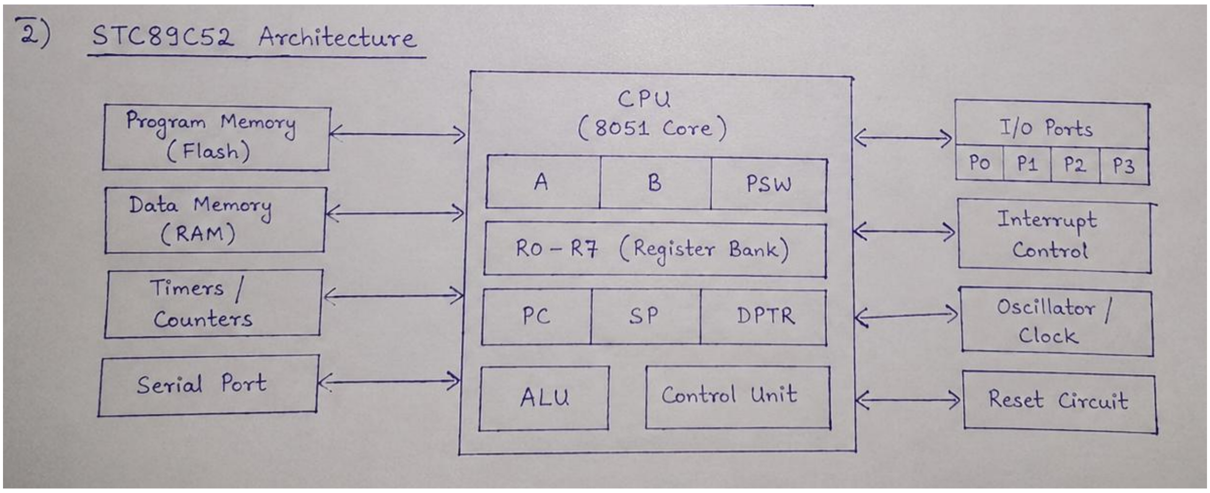
The interrupts can be **enabled, disabled, and prioritized** using special function registers such as **IE (Interrupt Enable)** and **IP (Interrupt Priority)**.

Simple Flow:

Main Program → Interrupt Occurs → Identify Source → Save State → Execute ISR → Restore State → Resume Main Program

Oscillator and Clock: The oscillator and clock system provides the timing required for CPU and peripheral operations. The timing system keeps instruction processing and peripheral activities synchronized.

STC89C52 Architecture.



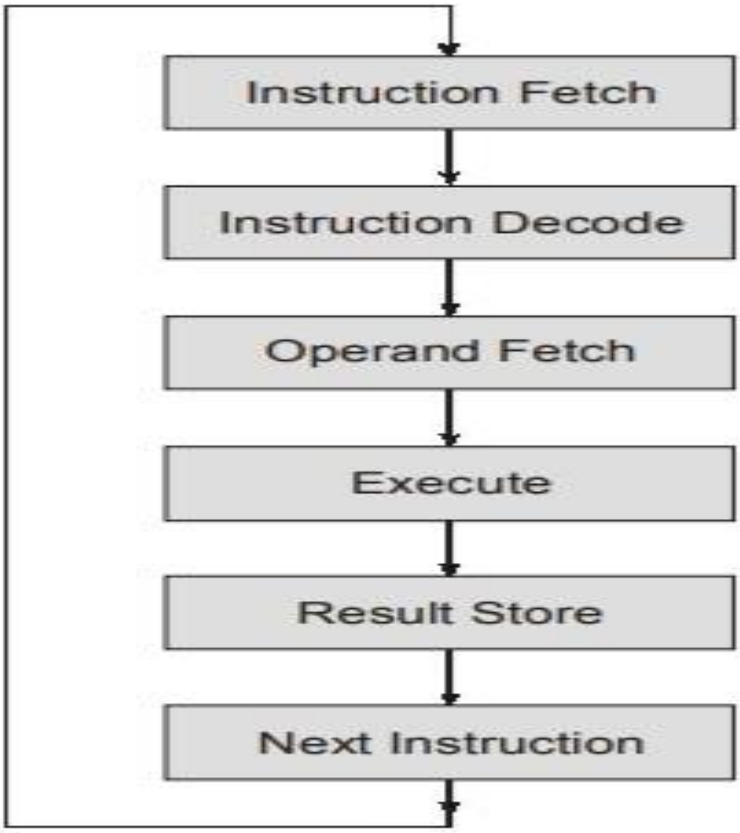
STC89C52 Instruction Set

The instruction set defines the operations that CPU can performs.

Category	Examples	Purpose
Data Transfer	MOV, PUSH, POP	Move data between registers, memory and other locations.
Arithmetic	ADD, ADDC, SUBB, INC, DEC, MUL, DIV	Perform numerical operations.
Logical	ANL, ORL, XRL, CLR, CPL	Perform logical and bit-related operations.
Branch / Control	SJMP, LJMP, AJMP, JZ, JNZ, DJNZ	Change the normal sequence of instruction execution.
Bit Operations	SETB, CLR, CPL and related bit instructions	Operate on individual bits.

Basic Instruction Execution.

Instruction execution describes how the CPU processes one instruction at a time. A simplified educational model can represent the process using four stages: Fetch, Decode, Execute and Write Back. The exact internal timing of an 8051-compatible processor can be more detailed, but this simplified model is useful for an educational simulator.



Instruction Execution Cycle

Fetch: Retrieve the next instruction from memory into the CPU.

Decode: Interpret the fetched instruction to determine the required action.

Execute: Perform the operation specified by the instruction.

Write Back: Store the result of execution into a register or memory.

Examples:

Ex 1: MOV A, #25H

Before execution: A = 00H

Execution:

PC

↓

Fetch: MOV A, #25H

↓

Decode: Data transfer instruction

↓

Execute: Load 25H into A

↓

A = 25H

Result: A = 25H

GPIO(General Purpose Input/Output)

GPIO pins are the pins of a microcontroller that can be programmed to either

Input → receive information from an external device.

Output → send a signal to an external device.

GPIO in STC89C52

The **STC89C52** has four main I/O ports:

Port	Pins	Number of pins
Port 0	P0.0 – P0.7	8
Port 1	P1.0 – P1.7	8
Port 2	P2.0 – P2.7	8
Port 3	P3.0 – P3.7	8

GPIO as Input

Button-----P1.0------Microcontroller

Button not pressed---1

Button Pressed----0

GPIO as Output

Microcontroller---P1.0----LED

P1.0=1---high

P1.0=0----Low.

Decisions:

- To implement FCFS CPU Scheduling to demonstrate how processes are selected and executed based on their arrival order.
- To integrate Process Management with the Microcontroller CPU process will be managed using PCB and a ready Queue ,the sent CPU for execution.
- To make CPU and memory operations visible in the simulator the user will be able to see the current instruction, registers ,pc ,memory address/value.